

LISA Response



Sun



$$L = 2.5 \cdot 10^6 \text{ km}$$



Key Differences w/ LIGO:

- LVK interferometers are independent.
LISA three coupled "interferometers".
- LVK is always in the long-wavelength limit.
+ static (non-moving) detector limit
LISA: high f , no longer in long-wavelength limit
+ detector motion is always important.
- LVK: does real interferometry
LISA: interferometry is done in software



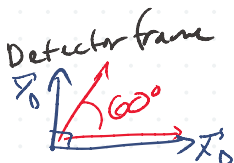
Measure phase shift of laser due to GWS.

Basic idea: to measure the relative phases of the laser light at each end of each arm.

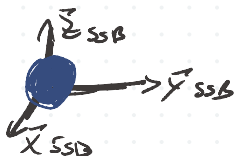
"Time delay interferometry"

Doppler tracking

Source frame $\rightarrow$ Solar-system Barycenter Frame (SSB) $\rightarrow$ Detector frame.



θ : ecliptic latitude
 ϕ : " longitude



Wave frame
 $\{\hat{e}, \hat{j}, \hat{k}\}$

$$\hat{e} = \cos\theta \cos\phi \hat{x}_{ssB} + \cos\theta \sin\phi \hat{y}_{ssB} - \sin\theta \hat{z}_{ssB}$$

$$\hat{j} = \sin\phi \hat{x}_{ssB} - \cos\phi \hat{y}_{ssB}$$

$$\hat{k} = \sin\theta \cos\phi \hat{x}_{ssB} + \sin\theta \sin\phi \hat{y}_{ssB} + \cos\theta \hat{z}_{ssB}$$

$$e^+ = \hat{e} \otimes \hat{e} - \hat{j} \otimes \hat{j}$$

$$e^x = \hat{i} \otimes \hat{j} + \hat{j} \otimes \hat{i}$$

$$h(\xi, \hat{k}) = h_+(\xi, \hat{k}) e^+ + h_x(\xi, \hat{k}) e^x$$

$$\xi = t + \hat{k} \cdot \vec{x}_D$$

$$e^+ = e^+ \cos 2\gamma - e^x \sin 2\gamma$$

$$e^x = e^+ \sin 2\gamma + e^x \cos 2\gamma$$

Detector tensor

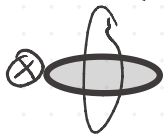
$$h = D_{ij} h^{ij} \quad (\text{LIGO})$$

$$D \equiv \frac{1}{2} (\hat{x} \otimes \hat{x}_D - \hat{y}_D \otimes \hat{y}_D)$$

LISA

$$h_{ij} = \int D_{ij} h^{ij}(\xi(\lambda)) d\lambda$$

$$= \frac{1}{2} \frac{\hat{r}_{ij}(t_i) \otimes \hat{r}_{ij}(t_j)}{1 - \hat{k} \cdot \hat{r}_{ij}(t_i)} \int_{\xi_i}^{\xi_j} h(\xi) d\xi$$



Define $H(a, b) \equiv \int_a^b h(\xi) d\xi$

$$h(\xi) = \int \tilde{h}(\omega) e^{i\omega\xi} d\omega$$

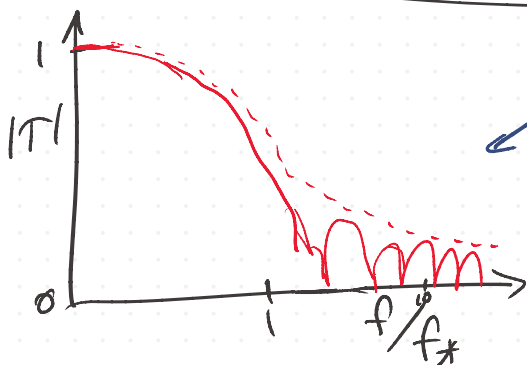
$$H(a, b) = \int \frac{\tilde{h}(\omega)}{i\omega} (e^{i\omega a} - e^{i\omega b}) d\omega.$$

$$h_{ij} = \int_{-\infty}^{\infty} \frac{1}{2} (\hat{r}_{ij}(t_i) \otimes \hat{r}_{ij}(t_i)) \mathcal{T}(f, t_i, \hat{k}) df$$

"LISA Transfer function"

$$\mathcal{T}(f, t, \hat{k}) = \text{sinc}\left(\frac{f}{2f_*} (1 - \hat{k} \cdot \hat{r}_{ij}(t))\right) e^{i\left(\frac{f}{2f_*} (1 - \hat{k} \cdot \hat{r}_{ij}(t))\right)}$$

$$f_* = \frac{1}{2\pi L}$$



Corrections compared to LVK from breaking $\lambda_{ow} \gg L$

Next class: either PTAs response or
- DATA ANALYSIS